

Patent Infringement Analysis — US 9,824,579 B2 (Samsung / Patkar): Method and Electronic Device for Selecting and Controlling a Home Network Device

TL;DR

- **No commercial product identified worldwide that literally infringes US 9,824,579 B2** — the patent's core element (a light/laser beam emitted by the controller and detected by a photo-sensor on the target HND, which then transmits its ID back) is not present in any shipping smart-home product we located. Even Samsung itself does not appear to have commercialized the method.
- **The strongest infringement risks are under the doctrine of equivalents (DoE) against UWB "point-to-control" products** — most notably Xiaomi's MIX4 "Point to connect" (NXP Trimension UWB, announced September 26, 2021) and the Sevenhugs Smart Remote X (Qorvo / 7Hugs Labs, launched 2018), which perform the same function (point controller at HND → controller auto-displays a per-device control UI) by substituting UWB time-of-flight ranging for the light-beam mechanism. Whether DoE survives is highly uncertain because the patent specification, claim 5, and the prosecution history all anchor the invention to *light*, and a court applying the "function-way-result" test would likely find the "way" (radio time-of-flight vs. directed photon beam detected by an on-device sensor) materially different.
- **All other candidates (Apple HomePod handoff, Samsung SmartThings AR Find / Quick Remote, Apple Vision Pro Home, AR camera-based remote apps, LiFi/VLC research systems) are LOW or NONE severity** — they either lack directed pointing, lack a per-HND sensor receiving an identifying beam, or lack the auto-activating per-device UI tied to a beam-detection event. The patent is also encumbered by acknowledged prior art ("Laser Guided Appliance Control System" 2012; IEEE 2011 paper) that may limit its effective scope.

Key Findings

1. Patent scope is narrow and light-anchored. Independent claims 1, 6, 7 and 12 of US 9,824,579 B2 all recite a "light signal" / "light beam event"; dependent claims 5 and 11 narrow the light source to a laser. The specification's described embodiments are a smartphone with an integrated laser pointer aimed at an HND that carries a photodiode/laser sensor; the HND responds over the home network with its identifier, which causes the controller app to auto-render a control UI specific to that HND. Samsung disclosed two prior-art references during prosecution — "Laser Guided Appliance Control System" (2012) and an IEEE paper from 2011 — meaning the claims were allowed only over those earlier light-pointing teachings. The novelty Samsung successfully argued is therefore quite specific (the registration-of-controller + beam-detection-event subscription + auto-activated UI workflow), and that narrow scope tightly constrains the range of equivalents available. (uspto) (uspto)

2. No literal infringer found. Across global smart-home platforms (Google Home/Nest, Amazon Alexa, Apple HomeKit/Vision Pro, Samsung SmartThings, Xiaomi Mi Home/Aqara, IKEA DIRIGERA, Philips Hue, LG ThinQ, Huawei HarmonyOS, Tuya, Bosch, Hubitat, Home Assistant, IFTTT) we found no shipping product that (a) emits a directed light/laser beam from the controller, (b) requires the target HND to carry a light/laser sensor that detects that beam, and (c) uses that beam-detection event to identify the HND back to the controller. Samsung's own SmartThings "Quick Remote" pop-up is proximity-triggered (BLE/Wi-Fi), not beam-triggered. Samsung's UWB-AR "find devices" feature in SmartThings uses UWB ranging + camera AR, not a light beam detected by a sensor on the appliance.

3. UWB "point-to-control" is the closest functional analogue — but architecturally distinct. Xiaomi's "Point to connect" feature in the MIX4 smartphone, powered by NXP's Trimension UWB chipset (announced September 26, 2021), explicitly matches the *function/result* of the patent — point the phone at a smart device and the phone "display[s] operation control cards on the phone screen to achieve instant connection." The Sevenhugs Smart Remote X (2018, \$300; 7Hugs Labs was acquired by Qorvo on November 5, 2020 for \$48.7 million per Qorvo's GlobeNewswire press release) likewise displays "dedicated interface to control each device you point at." However, both use **ultra-wideband radio time-of-flight ranging**, not light beams; neither target device contains a "light/laser sensor" detecting an emitted beam. These are doctrine-of-equivalents candidates only, and arguments against equivalence are strong (different physical phenomenon, different sensor type, different "way"). (NXP + 3)

4. Most AR/camera-based "point at device" apps avoid the claim entirely by using computer vision (recognizing the device in the camera image) and spatial anchoring rather than a beam emitted *from* the controller *to* a sensor *on* the HND. Apple Vision Pro's smart-home control uses eye-gaze + pinch on virtual UI overlaid in 3D space, not a light beam directed at and sensed by the appliance.

5. LiFi/VLC products (Signify Trulifi, pureLiFi, Oledcomm) communicate via modulated light but the topology is inverted from the patent: the *light source* is the LiFi luminaire (which sends to a USB key on a laptop), and the use case is broadband data transfer, not pointing a handheld controller at a target appliance to summon a control UI. They are not infringement candidates.

6. Academic and maker projects show a small "literal-style" footprint (e.g., Hackster.io "Hand Mounted Laser Controller" by Devon Palma; LaserCat IoT projects). These map onto the claims more closely than any commercial product, but they are non-commercial demonstrations with negligible enforcement value. (Hackster.io)

Details

A. Patent Recap and Claim Anchors

US 9,824,579 B2 (filed Oct. 7, 2015; priority Indian application 5021/CHE/2014 of Oct. 7, 2014; granted Nov. 21, 2017) recites four independent claims:

- **Claim 1 (method @ controller):** generate light signal → obtain ID from HND that detected light-beam event → activate control UI based on ID → send control messages through UI.
- **Claim 6 (method @ HND):** detect light-beam event → send HND's ID to controller → receive control info through auto-activated UI.
- **Claim 7 (controller apparatus):** controller + light source configured as in claim 1.
- **Claim 12 (HND apparatus):** controller configured as in claim 6.

Claims 2-4 (and 8-10) add dependent flavors: subscribing to a beam-detection event, receiving the ID via a gateway, or via broadcast. Claims 5 and 11 narrow the light source to a laser. (uspto)

(uspto)

B. Candidate-by-Candidate Analysis

B.1 Xiaomi MIX4 / "Point to connect" (NXP Trimension UWB)

- **What it does (per NXP GlobeNewswire press release, Sept. 26, 2021, quoting Rafael Sotomayor, Executive Vice President and General Manager of Connectivity and Security at NXP Semiconductors):** "the MIX4 smartphone can point to compatible Xiaomi devices and display operation control cards on the phone screen to achieve instant connection." Compatible HNDs include Xiaomi Sound smart speakers and TVs. Sotomayor added: "The groundbreaking MIX4 and its associated home device ecosystem demonstrate Xiaomi's ability to expand the boundaries of the smart home and pave the way for expanded UWB use cases." (NXP Semiconductors + 3)
- **How it works:** UWB chipsets (NXP SR100T family) on both ends; time-of-flight ranging + angle-of-arrival; no photonic sensor on the HND. Per Xiaomi's October 12, 2020 demo, pointing the phone causes "the phone's UI [to] automatically switch[] between various device control menus as the executive points at different smart home gadgets — no need to manually swipe to a specific device menu." (GlobeNewswire) (Android Authority)
- **Claim mapping:**
 - Claim 1(a) "generate a light signal from a light source": **NOT met literally** — UWB is a radio-frequency signal in 3.1-10.6 GHz, not light. DoE: arguably analogous (electromagnetic emission directed at the target), but the patent's narrow "light source" wording and the dependent "laser" claim cut against this.
 - Claim 1(b) "obtaining ID from HND detecting a light beam event": **NOT met** — the HND does not "detect a light beam"; UWB is bidirectional ranging.
 - Claim 1(c) "activating control UI based on ID": **MET** functionally — the MIX4 auto-pops the per-device control card.
 - Claim 1(d) "sending control messages through control UI": **MET**.
- **Literal infringement:** No (claim 1(a)/(b) not satisfied).
- **Doctrine of equivalents:** **Possible but weak.** Function (identify pointed-at device → show its UI) is similar; Way (UWB radio time-of-flight vs. directed photon beam + photo-sensor) is meaningfully different; Result (correct UI appears) is similar. Prosecution-history

estoppel may also apply if Samsung argued during examination that "light signal" distinguished it from radio-based prior art (not confirmed without the file wrapper).

- **Severity / Confidence: MEDIUM-LOW / Medium.**

B.2 Sevenhugs Smart Remote X (Qorvo / 7Hugs Labs, launched 2018)

- **What it does:** "When you point Smart Remote at any device, it automatically displays the right controls" (Sevenhugs Kickstarter; IDA Awards). Per TechHive (Michael Brown, Nov 10, 2016), "The Smart Remote already works with more than 25,000 Wi-Fi, Bluetooth, and infrared devices, according to Tchedikian. That list includes Samsung Smart TVs, Philips Hue and LIFX smart bulbs, Sonos speakers, and the Nest Learning Thermostat." [Kickstarter](#) [TechHive](#)
- **How it works (clarified by targeted subagent research):** The remote is positioned via three UWB anchors mounted in the room; the remote itself contains an IMU + UWB radio for time-of-flight triangulation. The user "teaches" each target device's coordinates during setup by physically holding the remote next to the device and pressing Locate (Sevenhugs support page: *"hold it at most 6 inches from whatever you wish to locate, and confirm. The device is now located and available for control when Smart Remote is in Point Mode"*). Outbound control commands then go via IR/Wi-Fi/Bluetooth. Per TechHive (Michael Brown, Nov 10, 2016), "Tchedikian says the Smart Remote can determine where it's located in the room within five to forty centimeters of precision." Critically: **no photodetector exists on the target appliance.** [TechHive + 2](#)
- **Claim mapping:** Identical analysis to Xiaomi — no literal infringement of the "light signal" / "light beam event" elements. Importantly, the target device does **not** detect any incoming signal whatsoever from the remote during selection; the remote infers what is being pointed at purely from its own position + orientation. This is even further from the claim language than Xiaomi (where the HND does at least respond to a UWB ping).
- **Literal infringement: No.**
- **Doctrine of equivalents: Weak.** The remote does not even send an identifying signal to the device — selection is entirely controller-side. Claim 1(b)'s "obtaining identification information from the HND detecting a light beam event" has no analogue here.
- **Severity / Confidence: LOW / High.**

B.3 Apple iPhone + HomePod (mini, 2nd gen) "Handoff" via U1/U2 UWB

- **What it does:** Bringing an iPhone (11 or later) "close to HomePod" auto-surfaces media controls and song suggestions on the iPhone; UWB enhances proximity Handoff and AirDrop prioritization. Apple's US Patent 11,044,405 ("Location systems for electronic device interactions with environment"), reported by iDownloadBlog (Christian Zibreg, June 22, 2021), describes that "an iPhone could display control icons when pointed at a supported piece of electronic equipment" — but as of May 2026 Apple has *not* shipped a feature that auto-activates a per-HomeKit-device UI on the iPhone purely based on directional pointing. [Engadget](#) [iDownloadBlog](#)
- **Claim mapping:**

- "Light signal" from controller: **NOT met** (UWB, not light).
- Auto-activated UI: HomePod handoff does show playback controls upon proximity, but this is a HomePod-specific media UI, not the generic per-HND control UI of the patent.
- **Literal: No. DoE: Weak** (same UWB-vs-light distinction; the feature is also proximity-rather than pointing-driven in shipping form).
- **Severity / Confidence: LOW / High.**

B.4 Samsung SmartThings — Galaxy Note20 Ultra "AR find/control" (UWB)

- **What it does (August 2020 update):** Galaxy users could "locate your compatible IoT devices and even interact with them through an AR interface" using the Note20 Ultra's UWB chip. (Android Police)
- **Claim mapping:** UWB ranging + camera-based AR. Not a light beam.
- **Special status:** Samsung is the **assignee** of the patent; any internal Samsung implementation is licensed by definition. No infringement risk from Samsung against itself.
- **Severity / Confidence: NONE / Very High.**

B.5 Apple Vision Pro + Home app (visionOS, 2024–2026)

- **What it does:** Vision Pro is a spatial computer where the user looks at virtual UI tiles floating in 3D space and pinches to confirm. Smart-home control happens through the Home app and HomeKit; third-party apps like Homerise and HomeUI overlay controls on physical room positions. (Apple Support)
- **Claim mapping:**
 - No light beam emitted toward the HND.
 - No sensor on the HND detecting a controller-emitted beam.
 - Eye gaze is sensed *internally* by the Vision Pro's IR cameras, not by the HND. (AppleMagazine)
- **Literal: No. DoE: No** (the "way" is eye-gaze + pinch on virtual UI; the HND is not involved in selection at all).
- **Severity / Confidence: NONE / Very High.**

B.6 AR camera-based remote-control apps and patents (Hayo; Aqara "Smart AR Home"; patents like US 11,435,891, US 9,900,541, US 11,163,434, US 11,868,582)

- **What they do:** Use the phone camera to recognize the target device or anchor a virtual control to a fixed spatial location; surface a per-device control UI when the user aims the camera.
- **Claim mapping:**
 - No light beam *emitted from* the controller toward the HND.

- Selection is computer-vision-based or AR-anchor-based, not photo-sensor-based on the HND.
- **Literal: No. DoE: Weak** — same function/result, but very different way (passive camera image processing vs. active light emission + on-device photodetector).
- **Severity / Confidence: LOW / High.**

B.7 LiFi / Visible Light Communication (Signify Trulifi, pureLiFi, Oledcomm; academic VLC smart-home papers, e.g. Xu et al., *Sensors* 22(21):8181, 2022)

- **What they do:** Trulifi modulates light from luminaires to deliver high-speed data to USB receivers on laptops/tablets. The 2022 MDPI *Sensors* paper (Xu et al.) describes a VLC + Bluetooth AC light-bulb that delivers 3D indoor localization for smart-home device control — interesting because the *smartphone* has a photodiode receiving VLC from ceiling lights, which is the inverse of the patent's controller-emits-light architecture.
- **Claim mapping:**
 - In Trulifi, the controller does not emit a directed light beam *toward* the HND.
 - In VLC indoor-positioning systems, light flows *from* room luminaires to the smartphone — opposite of claim 1(a).
- **Literal: No. DoE: No** (architectural direction is reversed).
- **Severity / Confidence: NONE-LOW / High.**

B.8 Hayo (augmented-reality system, 2017)

- Hayo turns the room into virtual buttons recognized by a stationary camera unit; gestures (no laser) control IoT. Severity: **NONE**. Gadgets.in

B.9 Hobbyist / academic laser-pointing projects (Hackster.io "Hand Mounted Laser Controller" by Devon Palma; LaserCat IoT)

- These literally use a hand-mounted laser aimed at photoresistor targets to trigger smart-plug/light actions over a network. Architecturally these are very close to the patent's literal teaching: laser at controller, photoresistor at target, IoT control follows.
- However, the auto-activation of a *per-HND control UI on the controller* (claims 1(c), 6, 7) is typically absent — the controller is dumb (the laser hand-mount has no screen / UI), and the target's photoresistor triggers a hardcoded action.
- **Literal: Partial / borderline** — depends on the specific build.
- **Severity / Confidence: LOW / Medium** (low commercial impact, low enforcement value, but a closer literal mapping than any major-vendor product).

B.10 Google Home / Nest, Amazon Alexa / Echo, Apple HomeKit (baseline), LG ThinQ, Huawei HiLink, Tuya, Bosch Smart Home, Hubitat, Home Assistant, IFTTT, Logitech Harmony, Caavo, Neco, Philips Hue, IKEA DIRIGERA

- None of these platforms ship a feature that uses controller-emitted light + HND-side photodetector + auto-UI activation. They use Wi-Fi/Zigbee/Thread/Matter/IR/BT/UWB for selection and control. **Severity: NONE.**

C. Patent-Validity Headwind

The '579 patent itself cites as prior art a 2012 "Laser Guided Appliance Control System" and a 2011 IEEE paper teaching light-based device control. Any litigation campaign would therefore have to articulate exactly which configurations were *not* taught by those references. The allowed claims hinge on the registration / subscription-to-beam-detection-event flow that auto-activates a device-specific UI — features that any defendant could attack on §103 obviousness grounds in IPR.

D. Severity Ranking Table

Rank	Product / System	Mechanism	Literal?	DoE?	Severity	Confidence
1	Xiaomi MIX4 "Point to connect" (NXP Trimension UWB)	UWB time-of-flight + auto control UI	No	Possible	MEDIUM-LOW	Medium
2	Sevenhugs Smart Remote X (Qorvo / 7Hugs Labs)	UWB anchors + IMU triangulation + per-device UI	No	Weak	LOW	High
3	Hobbyist laser-pointer IoT projects (Hackster, etc.)	Laser + photoresistor + smart-plug	Borderline	n/a	LOW	Medium
4	AR camera / AR-anchor remote apps (Hayo, US 11,435,891, etc.)	Camera CV + anchor + UI	No	Weak	LOW	High
5	Apple iPhone + HomePod UWB Handoff	UWB proximity-triggered media UI	No	Weak	LOW	High
6	Apple Vision Pro Home app / HomeUI / Homerise	Eye-gaze + pinch on virtual UI	No	No	NONE	Very High

Rank	Product / System	Mechanism	Literal?	DoE?	Severity	Confidence
7	Samsung SmartThings AR / Find / Quick Remote	UWB / BLE proximity / camera AR	No	Moot (Samsung-owned)	NONE	Very High
8	LiFi / Trulifi / VLC smart-home research	Modulated light luminaire → USB key	No (direction inverted)	No	NONE	High
9	Google Nest / Alexa / HomeKit / LG ThinQ / Tuya / Hue / DIRIGERA / etc.	Wi-Fi / Zigbee / Thread / Matter app control	No	No	NONE	Very High

Recommendations

Stage 1 — Validity audit first (highest leverage): Before sending any pre-suit letter, commission an independent prior-art search focused on light-pointing device selection between 2008–2014, paying particular attention to the cited "Laser Guided Appliance Control System" (2012) reference and the IEEE 2011 paper. Pull the file wrapper for US 9,824,579 and the Indian priority application 5021/CHE/2014 to map the exact prosecution-history estoppel boundary around the "light signal" element. *Threshold to abandon:* if §102/§103 art clearly shows the "registered controller + beam-detection-event subscription + auto-UI activation" combination predating Oct. 7, 2014, drop the campaign.

Stage 2 — Targeted DoE study on Xiaomi MIX4 "Point to connect": This is the only commercially shipping product whose user-visible function is materially indistinguishable from the patent's promise (point phone at smart device → per-device UI appears → control). Engage an outside expert to construct a function-way-result analysis and a "vitiation" defense rebuttal (i.e., would substituting UWB for "light signal" vitiate the claim term?). *Trigger for action:* obtain a working teardown sample of a MIX4 + UWB-enabled Xiaomi appliance, confirm the UI activation handshake, and benchmark against claim 1's four steps.

Stage 3 — Defensive watch on Apple, Google, Samsung-licensee filings: Apple's US Patent 11,044,405 on UWB-pointed iPhone control of accessories signals that the U2/HomePod ecosystem may add literal "point-to-control" UI features. Set a recurring (quarterly) watch on USPTO filings citing the '579 patent, on FCC certifications for smart appliances claiming light/laser sensors, and on Samsung internal licensing disclosures. *Threshold to escalate:* a U.S.-shipping product that (a) emits an explicit directed light/optical beam from a controller and (b) places a photo-sensor on the appliance.

Stage 4 — Do not pursue: Sevenhugs/Qorvo (selection is purely controller-side, the HND has no detector), Apple Vision Pro, all Wi-Fi/Zigbee/Matter platforms, LiFi/VLC products, and

Samsung's own SmartThings (assignee-owned). The cost-benefit of asserting against these is negative.

Stage 5 — Licensing posture: If validity holds and Stage 2 yields a clean DoE theory, the realistic monetization path is a low-rate license to UWB "point-to-control" implementers (Xiaomi, Qorvo's UWB licensees, and any future Google Pixel or Apple UWB-point feature). Do not posture for an injunction — the doctrine-of-equivalents posture and the prior-art cloud make injunctive relief unlikely.

Caveats

- **This is a technical infringement analysis, not a legal opinion.** Final infringement determinations turn on claim construction (Markman) and the prosecution-history file wrapper, which has not been pulled for this study.
- **Doctrine-of-equivalents analysis is inherently uncertain.** Courts vary in their willingness to extend "light signal" to cover "UWB radio signal," and prosecution-history estoppel could foreclose the argument entirely. *Festo* and *Warner-Jenkinson* considerations apply.
- **Samsung-as-assignee.** Samsung may have implicitly licensed Tizen partners, FiRa Consortium co-members (Apple, Xiaomi, NXP, Qorvo), or other ecosystem partners under cross-license agreements that are not publicly visible. A pre-suit due-diligence sweep of Samsung's published license registry and FRAND commitments (if any) is essential before sending letters.
- **Patent validity not independently verified.** The '579 patent's own admitted prior art is concerning; an IPR challenge by a deep-pocketed defendant could cancel the claims.
- **Some matches (e.g., Apple "point-to-control" UWB) are based on patent filings rather than shipping features.** Those are research/prototype signals only and should not be treated as commercial infringement until launched.
- **Search coverage limited to public sources.** Internal corporate documentation, FCC datasheets behind paywalls, and non-English (Chinese, Korean, Japanese) trade-press coverage may surface additional candidates; a follow-on multilingual sweep is recommended before pre-suit action.
- **No active litigation citing US 9,824,579 was identified** in the public record as of May 21, 2026. Samsung has not, to our knowledge, asserted the patent against any party.